

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Mathematics Extension 2

HSC TASK 2

JUNE 2007

TIME ALLOWED: 70 minutes

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.

(FOR MARKERS USE ONLY)

Q1	Q2	Q3	TOTAL
/20	/20	/20	/60

QUESTION 1: (20 MARKS)

Marks

3 (a) Find $\int \frac{2x}{\sqrt{x+1}} dx$

2 (b) Find $\int \frac{dx}{\sqrt{4x-x^2}}$

3 (c) Find $\int x^2 e^x dx$

4 (d) Evaluate $\int_0^{\pi} \sin^3 x dx$

3 (e) Find $\int_0^{\frac{\pi}{2}} \frac{1}{1+\sin \theta} d\theta$

3 (f) (i) Find values of A, B and C for which

$$A(x+1)^2 + B(x-1) + C(x+1)(x-1) = 8x-4$$

and hence, or otherwise, express $\frac{8x-4}{(x-1)(x+1)^2}$ in the form

$$\frac{A}{x-1} + \frac{B}{(x+1)^2} + \frac{C}{x+1}$$

2 (ii) Using your solution to (i) above, find

$$\int \frac{8x-4}{(x-1)(x+1)^2} dx$$

QUESTION 2: (20 MARKS)

Marks

- (a) For the hyperbola $\frac{x^2}{2} - \frac{y^2}{2} = 1$ find
- 2 (i) The foci
- 2 (ii) The equations of the directrices
- 1 (iii) The equations of the asymptotes
- 1 (iv) The length of the major axis
- (b) The point $P(3t, \frac{3}{t})$ lies on the hyperbola $xy = 9$
- 3 (i) Prove that P is the midpoint of the line MN where M and N are the points where the tangent to the hyperbola at P cuts the x and y axes respectively.
- 2 (ii) Show that the midpoint of PM lies on another hyperbola and give its equation
- 4 (c) (i) **Derive** the equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point P $(asec\theta, btan\theta)$
- Give your answer in the form $Ax + By = 1$
- 3 (ii) The tangents at the points P $(asec\theta, btan\theta)$ and Q $(asec\alpha, btan\alpha)$ meet at right angles.
- Prove that $\sin\theta\sin\alpha = -\frac{b^2}{a^2}$
- 2 (d) P is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose foci are S and S'.
- Use the focus-directrix definition of the hyperbola to prove that $|PS - PS'|$ is a constant

QUESTION 3: (20 MARKS)

Marks

- 3 (a) Find the values of a and b so that 2 is a double root of the polynomial

$$x^4 + ax^3 - 3x^2 - bx + 4 = 0$$

- 2 (b) (i) Find (in expanded form) the equation whose roots exceed by 1 the roots of

$$x^3 + 6x^2 - 3x + 1 = 0$$

- 4 (ii) If α , β and γ are the roots of the polynomial $x^3 - 2x^2 + 3x - 4$
prove that $\beta^2\gamma^2 + \gamma^2\alpha^2 + \alpha^2\beta^2 = -7$.

Show that the polynomial whose roots are $\beta^2\gamma^2$, $\gamma^2\alpha^2$ and $\alpha^2\beta^2$
is $x^3 + 7x^2 - 32x - 256$

- 4 (c) Solve the quartic equation $x^4 + 2x^3 + x^2 - 1 = 0$ given that one root is
 $-\frac{1}{2} + \frac{i\sqrt{3}}{2}$

- 3 (d) (i) Using deMoivre's Theorem, or otherwise, obtain an expression for $\cos 4\theta$ in terms of $\cos \theta$

(NOTE: $(x+a)^4 = x^4 + 4x^3a + 6x^2a^2 + 4xa^3 + a^4$)

- 4 (ii) By considering the roots of the equation $16x^4 - 16x^2 + 1 = 0$ and using the substitution $x = \cos \theta$ and your answer to part (i) above,

show that $\cos \frac{\pi}{12} \cos \frac{5\pi}{12} \cos \frac{7\pi}{12} \cos \frac{11\pi}{12} = \frac{1}{16}$

11/10/20

$$P(u) = (u + \frac{1}{2} - \frac{i\sqrt{3}}{2})(u + \frac{1}{2} + \frac{i\sqrt{3}}{2})Q(u)$$

$$= [(u + \frac{1}{2})^2 + \frac{3}{4}]Q(u) \quad \leftarrow \textcircled{1}$$

$$= (u^2 + u + 1)Q(u) \quad \leftarrow \textcircled{1}$$

$$u^2 + u + 1 \mid \frac{u^2 + u - 1}{u^2 + 2u^3 + u^2} \quad \leftarrow \textcircled{1}$$

$$\frac{u^4 + u^3 + u^2}{u^2}$$

$$-1$$

$$\frac{u^3 + u^2 + u}{-u^2 - u - 1}$$

$$\frac{-u^3 - u - 1}{-u^3 - u - 1}$$

$$\frac{-u^3 - u - 1}{-u^3 - u - 1}$$

$$\text{Now } u^2 + u - 1 \Rightarrow u = \frac{-1 \pm \sqrt{5}}{2} \quad \leftarrow \textcircled{1}$$

$$= -\frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

$$\therefore \text{ Roots are } -\frac{1}{2} + \frac{i\sqrt{3}}{2}, -\frac{1}{2} - \frac{i\sqrt{3}}{2}, -\frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{1}{2} - \frac{\sqrt{5}}{2}$$

$$\therefore \text{ Sum of Roots is } -1 + \alpha + \beta = -2 \quad \left. \begin{matrix} \therefore \alpha + \beta = -1 \quad (1) \\ \end{matrix} \right\} \textcircled{1}$$

$$\text{Product is } (-\frac{1}{2} + \frac{i\sqrt{3}}{2})(-\frac{1}{2} - \frac{i\sqrt{3}}{2})\alpha\beta = -1$$

$$\therefore (\frac{1}{4} + \frac{3}{4})\alpha\beta = -1 \quad \left. \begin{matrix} \alpha\beta = -1 \quad (2) \\ \end{matrix} \right\} \textcircled{1}$$

$$\text{into (1)} \quad \alpha - \frac{1}{2} + 1 = 0$$

$$\alpha^2 - 1 + \alpha = 0$$

$$\alpha = \frac{-1 \pm \sqrt{5}}{2}$$

$$\therefore \text{ Roots are } -\frac{1}{2} \pm \frac{i\sqrt{3}}{2}, -\frac{1}{2} \pm \frac{\sqrt{5}}{2} \quad \leftarrow \textcircled{1}$$

$$d) \int_0^\pi \sin^3 x \, dx$$

$$= \int_0^\pi \sin^2 x \cdot \sin x \, dx$$

$$= \int_0^\pi (1 - \cos^2 x) \cdot \sin x \, dx$$

$$\text{Let } u = \cos x \quad x = \pi \quad u = -1$$

$$du = -\sin x \quad x = 0 \quad u = 1$$

$$= \int_1^{-1} (1 - u^2) \, du$$

$$= \int_1^{-1} (1 - u^2) \, du$$

$$= 2 \left[u - \frac{u^3}{3} \right]_1^{-1}$$

$$= 2 \left[\frac{2}{3} \right]$$

$$= \frac{4}{3}$$

$$= \frac{4}{3}$$

$$= \frac{4}{3}$$

$$= \frac{4}{3}$$

$$e) \int_0^{\pi/2} \frac{1}{1 + \sin \theta} \, d\theta$$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\frac{d\theta}{dt} = \frac{2}{1+t^2}$$

$$d\theta = \frac{2}{1+t^2} \, dt$$

$$\sin \theta = \frac{2t}{1+t^2}$$

$$\theta = \pi/2 \quad t = 1$$

$$\theta = 0 \quad t = 0$$

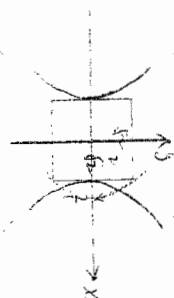
$$\theta = \pi/2 \quad t = 1$$

$$\theta = 0 \quad t = 0$$

$$\theta = \pi/2 \quad t = 1$$

$$\theta = 0 \quad t = 0$$

QUESTION 2



i) foci $(\pm c, 0)$

ii) $ae = 2$

$\therefore e = \sqrt{2}$

directrices at $x = \pm \frac{a}{e}$ i.e. $\pm \frac{\sqrt{2}}{2}$

$\therefore x = \pm 1$

iii) $y = \pm x$

iv) $2\sqrt{2}$

b) i) Eqn of tangent at P:

$x + t^2 y = 6t$

At M ($y = 0$) $x = 6t$

At N ($x = 0$) $y = \frac{6}{t}$

$\therefore$ Mid point is $\left(\frac{6t+0}{2}, \frac{\frac{6}{t}+0}{2} \right)$

i.e. $\left(3t, \frac{3}{t} \right)$ which is point P.

ii) $P(3t, \frac{3}{t}), M(6t, 0)$

Mid point of PM is $\left(\frac{9t}{2}, \frac{3}{2t} \right)$

i.e. If $X = \frac{9t}{2}$ & $Y = \frac{3}{2t}$ then $XY = \frac{9}{4}$

which is the equation of a hyperbola.

c) i)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Differentiating:

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\text{i.e. } \frac{dy}{dx} = \frac{b^2 x}{a^2 y} \text{ and at P}$$

$$m = \frac{b \sec \theta}{a \tan \theta}$$

So the equation of the tangent at P is:

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$\text{i.e. } \frac{y \tan \theta}{b} - \tan^2 \theta = \frac{x \sec \theta}{a} - \sec^2 \theta$$

$$\text{i.e. } \sec^2 \theta - \tan^2 \theta = \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b}$$

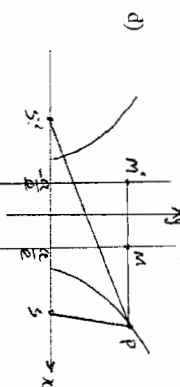
$$\text{i.e. } \frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

ii) Since tangents are perpendicular:

$$\frac{b \sec \theta}{a \tan \theta} \times \frac{b \sec \alpha}{a \tan \alpha} = -1$$

$$\text{i.e. } \frac{b^2}{a^2} \frac{1}{\cos \theta \cos \alpha} = -1 \Rightarrow \frac{\sin \theta \sin \alpha}{\cos \theta \cos \alpha}$$

$$\text{i.e. } \sin \theta \sin \alpha = -\frac{b^2}{a^2}$$



$$|PS - PS'| = |ePM - ePM'|$$

$$= e|PM - PM'|$$

$$= 2a \text{ which is constant}$$

$$(3) (a) P(x) = x^4 + ax^3 - 3x^2 - 5x + 4 = 0$$

$$P'(x) = 4x^3 - 6x - b$$

$$P(2) = 16 + 8a - 12 - 10 + 4 = 0 \quad (1)$$

$$\Rightarrow 8a - 2b + 8 = 0$$

$$4a - b + 4 = 0 \quad (1)$$

$$P'(2) = 32 + 12a - b = 0 \quad (2)$$

$$12a - b + 20 = 0 \quad (2)$$

$$(a) - (1) \quad 8a + 16 = 0$$

$$a = -2$$

$$(b) (i) P(x-1) = (x-1)^3 + 6(x-1)^2 - 3(x-1) + 1 \quad (1)$$

$$\therefore Q(x) = x^3 - 3x^2 + 3x - 1 + 6x^2 - 12x + 6 - 3x + 3 + 1$$

$$Q(x) = x^3 + 3x^2 - 12x + 9 \quad (1)$$

$$(ii) (\alpha\beta + \alpha\gamma + \beta\gamma)^2 = \alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 + 2\alpha\beta\gamma + 2\alpha\beta^2\gamma + 2\alpha\gamma^2\beta + 2\beta^2\gamma^2 + 2\alpha\gamma\beta^2 + 2\alpha\beta\gamma^2 + 2\beta\gamma\alpha^2 + 2\gamma\alpha\beta^2 + 2\gamma\beta\alpha^2 + 2\alpha\beta\gamma^2 + 2\alpha\gamma\beta^2 + 2\beta\gamma\alpha^2 + 2\gamma\alpha\beta^2 + 2\gamma\beta\alpha^2$$

$$\therefore \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(2 + \beta + \gamma)$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(4)(2)$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 16$$

$$= -7 \quad (1)$$

$$\text{Sum of Roots is } \beta^2\gamma^2 + \gamma^2\alpha^2 + \alpha^2\beta^2 = -7$$

$$\text{Sum of Roots} \times 2 \text{ is } \beta^2\gamma^2\alpha^2 + \alpha^2\beta^2\gamma^2 + \alpha^2\gamma^2\beta^2$$

$$= \beta^2\gamma^2\alpha^2(\gamma^2 + \beta^2 + \alpha^2)$$

$$= (4)^2 [(2)^2 - 2(3)]$$

$$= 16[-2]$$

$$= -32$$

$$\text{Product of Roots is } \alpha^4\beta^4\gamma^4$$

$$= (\alpha\beta\gamma)^4$$

$$= (4)^4$$

$$= 256$$

$$\therefore P(x) \text{ is } x^3 + 7x^2 - 32x - 256$$

$$(c) \text{ One root is } -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

$$\text{Another root is } -\frac{1}{2} - i\frac{\sqrt{3}}{2} \quad (1)$$

Question 1

$$a) \int \frac{2x}{\sqrt{x+1}} dx$$

$$\text{let } u^2 = x+1$$

$$\therefore 2u du = dx$$

$$= \int \frac{2(u^2-1) 2u du}{u}$$

$$= 4 \int u^2 - 1 du$$

$$= 4 \left[\frac{u^3}{3} - u \right]$$

$$= 4 \left[\left(\frac{x+1}{3} \right)^{3/2} - (x+1)^{1/2} \right] + C$$

$$b) \int \frac{dx}{\sqrt{4x^2-22}} = \int \frac{dx}{\sqrt{4-(x^2-4x+4)}}$$

$$= \int \frac{dx}{\sqrt{4-(x-2)^2}}$$

$$= \sin^{-1} \left(\frac{x-2}{2} \right) + C$$

$$c) \int x^2 e^x dx = \int x^2 \frac{d(e^x)}{dx} dx$$

$$= x^2 e^x - \int 2x e^x$$

$$= x^2 e^x - 2 \int x \frac{d(e^x)}{dx} dx$$

$$= x^2 e^x - 2 \left[x e^x - \int e^x dx \right]$$

$$= x^2 e^x - 2x e^x + 2e^x + C$$

$$\begin{aligned}
 &= -2 \left[(t+1)^{-1} \right]_0^1 \\
 &= -2 \left[2^{-1} - 1^{-1} \right] \\
 &= -2 \left[\frac{1}{2} - 1 \right] \\
 &= 1
 \end{aligned}$$

$$f) i) A(x+1)^2 + B(x-1) + C(x+1)(x-1) = 8x-4$$

$$x=1 \quad 4A$$

$$A$$

$$x=-1 \quad -2B$$

$$B$$

Comparing coefficient of x^2

$$1+C$$

$$C$$

$$= 0$$

$$= -1$$

$$\therefore \frac{8x-4}{(x-1)(x+1)^2} = \frac{1}{x-1} + \frac{6}{(x+1)^2} + \frac{1}{x+1}$$

$$ii) \int \frac{8x-4}{(x-1)(x+1)^2} dx = \int \frac{1}{x-1} dx + \int \frac{6}{(x+1)^2} dx - \int \frac{1}{x+1} dx$$

$$= \ln(x-1) - \ln(x+1) - 6(x+1)^{-1} + C$$

$$= \ln \left[\frac{x-1}{x+1} \right] - \frac{6}{x+1} + C$$

$$(d) i) (\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6i^2 \cos^2 \theta \sin^2 \theta + 4i^3 \cos \theta \sin^3 \theta + i^4 \sin^4 \theta \quad (1)$$

$$\therefore \cos^4 \theta + i \sin^4 \theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta + 4i (\cos^3 \theta \sin \theta - \cos \theta \sin^3 \theta)$$

Comparing real parts,

$$\begin{aligned}
 \cos^4 \theta &= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta) \\
 &= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 + \cos^4 \theta - 2 \\
 &= 8 \cos^4 \theta - 8 \cos^2 \theta + 1 \quad (1)
 \end{aligned}$$

$$(ii) \text{ let } x = \cos \theta$$

$$\therefore 16 \cos^4 \theta - 16 \cos^2 \theta + 1 = 0$$

$$\therefore 2(8 \cos^4 \theta - 8 \cos^2 \theta + 1) - 1 = 0 \quad (1)$$

$$\therefore 2 \cos^4 \theta - 1 = 0$$

$$\cos^4 \theta = \frac{1}{2} \quad (1)$$

$$\therefore 4\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3} \quad \text{ONLY } 4\theta$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12} \quad (1)$$

$$\therefore x = \cos \theta = \cos \frac{\pi}{12}, \cos \frac{5\pi}{12}, \cos \frac{7\pi}{12}, \cos \frac{11\pi}{12}$$

$$\text{Product of Roots} = \frac{1}{16} \quad (1)$$

$$\therefore \cos \frac{\pi}{12} \cos \frac{5\pi}{12} \cos \frac{7\pi}{12} \cos \frac{11\pi}{12} = \frac{1}{16}$$